

Mohamed Ashfaq

Computer Science Undergraduate | Full-Stack & IoT Developer
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PROFILE

Full-Stack Developer and Computer Science undergraduate with a strong track record of independently designing, building, and deploying end-to-end web applications using **Java (Spring Boot), React, and PostgreSQL**. Experienced in **REST API design, JWT authentication, CI/CD pipelines**, and Agile/Scrum workflows — with **8 shipped projects** spanning full-stack web, IoT, desktop, and machine learning. Committed to writing clean, maintainable code and delivering production-ready software.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C++, HTML5, CSS3

Back-End / Databases: Spring Boot, MySQL, PostgreSQL, REST API, JWT Authentication, Microservices

Front-End Frameworks: React, Next.js, Flutter

DevOps & Tools: Docker, CI/CD (GitHub Actions), Git, GitHub, Version Control, Postman, Figma, Electron

Testing: Unit Testing, JUnit, H2 In-Memory Database, Automated Testing

Data & ML: Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Jupyter Notebooks

Core Concepts: OOP, DSA, Agile/Scrum, Distributed Systems, IoT, Machine Learning, Team Leadership

PROJECTS & EXPERIENCE

Airea — Smart Respiratory Monitor for Early Lung Cancer

Sep 2025 – Apr 2026

GitHub: <https://github.com/Siddiha/Airea> | Live: <https://airea.lk>

C++ (ESP32), Flutter, Spring Boot, Supabase, FCM | *Freelance Team Project*

- Engineered an **end-to-end IoT health monitoring platform** over **7 months** with a **4-member team** — covering ESP32 wearables, a **Spring Boot REST API** backend, and a Flutter app with Supabase storage and FCM real-time alerts.
- Followed **Agile/Scrum** methodology with fortnightly sprints and **version control** via Git/GitHub across hardware, backend, and mobile workstreams.
- Designed the full **system architecture** — bridging embedded C++ firmware on ESP32, a **Spring Boot microservice** backend, and a cross-platform Flutter mobile app into a single cohesive platform.

MediBook — Clinic Appointment System

Mar 2026 – Apr 2026

GitHub: <https://github.com/ashfaq-ui/MediBook> | Live: <https://medi-book-flax.vercel.app/>

React, Spring Boot, PostgreSQL, JWT, GitHub Actions, Vercel, Railway | *Freelance Project*

- Built **solo in under 6 weeks** — a full-stack clinic system with **3 user roles** (Patient, Doctor, Admin), **15+ REST API endpoints**, JWT authentication, BCrypt hashing, and Gmail SMTP notifications.
- Set up **CI/CD pipeline** via GitHub Actions with **automated unit testing** (H2 in-memory); deployed to Vercel and Railway — reducing manual deployment effort to zero.
- Built a **React** frontend with **protected routes**, role-specific dashboards, and responsive UI for all 3 user roles — connecting to the **Spring Boot REST API** with Axios.

Invoice Manager — Desktop Finance & Sales Ledger App

Mar 2026 – Apr 2026

GitHub: <https://github.com/ashfaq-ui/invoice-manager>

Electron, JavaScript, Node.js | *Freelance Project*

- Delivered **solo in 3 weeks** — cross-platform desktop app for Mac and Windows managing invoices, sales ledger, and financial analysis across **3 report types** with zero-install distribution via GitHub Releases.

Loan Approval Prediction — Machine Learning

Jan 2026 – Apr 2026

GitHub: <https://github.com/ashfaq-ui/loan-approval-ml>

Python, Scikit-learn, Pandas, NumPy, Matplotlib, Plotly, Jupyter | *Academic Project*

- Built an end-to-end **ML pipeline** on **58,640 samples**; KNN tuned via GridSearchCV achieved **92.21% accuracy** and **0.9519 AUC-ROC** — the best performing model across 3 classifiers.

- Performed full **data preprocessing** — outlier removal, missing value imputation, StandardScaler feature scaling — and implemented **Voting Ensemble** and pruned **Decision Tree Regressor** with feature importance analysis.

SpendWise — Distributed Full-Stack Expense Tracker

Feb 2026 – Mar 2026

GitHub: <https://github.com/ashfaq-ui/spend-wise> | Live: <https://jazzy-concha-e4ad62.netlify.app/>

Java (Spring Boot), React.js, PostgreSQL, REST API | Freelance Project

- Developed **solo in 4 weeks** — a 3-tier personal finance app with **10+ REST API endpoints**, PostgreSQL persistence, and a React frontend with real-time updates; deployed and live on Netlify.
- Designed a normalized **PostgreSQL schema** for transaction and category management; implemented full **CRUD operations** with **Spring Boot JPA** repositories.

Black Olives — Premium Apple Retail E-Commerce Platform

Jun 2025 – Sep 2025

GitHub: <https://github.com/ashfaq-ui/webpro1-nextjs> | Live: <https://web-pro-1.netlify.app/>

Next.js | Freelance Project

- Built **solo in 4 months** — a full **Next.js SSR** e-commerce platform with product catalogue, dynamic routing, and SEO-optimized pages; live and deployed.
- Leveraged **Next.js getServerSideProps** and dynamic routes to pre-render product pages — improving **page load performance** and search engine indexability.

EDUCATION

BSc (Hons) in Computer Science

Sep 2024 – Present

Informatics Institute of Technology (Affiliated with University of Westminster), Colombo

- Current Average: 76.8% | Notable: Computer System Fundamentals (97%), Mathematics for Computing (88%), Software Development II — Java (83%)
- Level 05 (In Progress): Database Systems (87%), OOP (78%), SDGP (81%), Machine Learning & Data Mining, Algorithms, Client Server Architecture

Primary & Secondary Education

Jan 2010 – Dec 2023

Hindu College, Rathmalana

- G.C.E. Advanced Level — Physical Science: Combined Mathematics & Physics.
- G.C.E. Ordinary Level: Strong results including distinction in Mathematics.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Hackathon Finalist — CodeRally 6.0 and Xtreme 19.0, focusing on rapid development and problem-solving under pressure.
- Competitor — Hult Prize competition and SLIIT MATHFEST 2023.
- IEEE Member at Informatics Institute of Technology (2024 – Present).
- Leadership — Prefect (2019–2022) and Event Coordinator at Hindu College, Rathmalana.

CERTIFICATIONS

- Learning Docker (LinkedIn Learning, 2026)
- TypeScript for JavaScript Developers (LinkedIn Learning, 2025)
- React Essential Training (LinkedIn Learning, 2025)
- Critical Thinking and Problem Solving (LinkedIn Learning, 2024)
- Python GUI Development with Tkinter | Learning Python (LinkedIn Learning, 2024)

REFERENCES

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